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ARTICLE

Factors associated with uncontrolled hypertension in Santander, Colombia: baseline findings from the RE-HOPE study

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In Latin America, hypertension prevalence varies widely, and control rates remain low, particularly in low-income communities, despite the proven efficacy of recommended interventions. We aimed to describe baseline characteristics and identify individual factors associated with uncontrolled hypertension in participants of the RE-HOPE study in Santander, Colombia. A cross-sectional analysis was conducted including participants ≥18 years old with hypertension, defined by self-reported history, antihypertensive use, or systolic BP ≥ 140 and/or diastolic BP ≥ 90 mmHg. Descriptive statistics were used for baseline characteristics. Mixed-effects logistic regression models stratified by sex were applied to identify factors associated with uncontrolled hypertension, adjusting for age, diabetes, obesity, medication intake, physical activity, and smoking. A total of 3 481 participants were included, with a mean age of 60.2 years (SD 13.7); 62% were female. Mean systolic and diastolic BP were 144 ± 16 mmHg and 86 ± 10 mmHg, respectively. Despite high levels of awareness (81%) and medication use (76%), only 26% of participants had controlled hypertension (< 140/90 mmHg), and 9.6% met the target below 130/80 mmHg. Control rates were higher in the capital city compared to peripheral and rural provinces. Being female was inversely associated with uncontrolled hypertension (OR = 0.62; 95% CI: 0.52–0.75). In contrast, age ≥ 60 years (OR = 2.19; 95% CI: 1.84–2.61) and current smoking (OR = 2.48; 95% CI: 1.54–4.00) were positively associated with poor control. This study highlights significant challenges in hypertension management, particularly early diagnosis and the identification of priority groups, and emphasizes the need for targeted public health initiatives and policy actions to enhance hypertension control.

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INTRODUCTION

Hypertension is a leading contributor to cardiovascular disease (CVD) worldwide [1]. By 2019, it was estimated that the global prevalence of hypertension in adults was 33% but only 26% was well controlled [2]. In Latin America, the reports of hypertension prevalence vary widely, ranging from 18 to 62% [3, 4]. Unlike other regions where the prevalence of hypertension has declined in recent years, the Caribbean and Latin America have not seen the same trend [4, 5]. Besides, control is low in both men (23%) and women (35%) [3]. The above is worrisome given the high burden of disease that hypertension carries. A sub-analysis of the Prospective Urban Rural Epidemiology (PURE) study in four South American countries (Argentina, Brazil, Chile, Colombia) estimated that hypertension had an 18.7% population attributable fraction for CVD [6]. In Colombia, a community-based study of adults between 35 and 70 years reported a prevalence and control of hypertension of 37.5 and 17.3%, respectively [7]. Later, the May Measurement Month 2019 campaign (n = 48 324) reported a comparatively higher hypertension control rate (38.4%) [8]. These low rates of control have been consistently reported despite the efficacy of pharmacological and non-pharmacological interventions being recommended by guidelines [7–10]. Although a community-based intervention has demonstrated encouraging results in controlling hypertension [11], and some initiatives have been proposed [12], more evidence is needed to guide effective interventions in Colombia's low-income urban communities and achieve the World Health Organization's (WHO) 2030 goal of a 25% reduction in uncontrolled hypertension [13].

Several individual, provider, and health system factors are associated with uncontrolled hypertension. Individual factors include socioeconomic and marital status, as well as lifestyle factors such as smoking, diet, salt intake, and obesity [14, 15]. In addition, there is a significant disparity in hypertension awareness, treatment, and control between high-income countries (HICs) and lower-middle-income countries (LMICs). This disparity is closely related to the social determinants of health, including socioeconomic status, education, gender inequalities, and access to healthcare services [7, 16]. Therefore, understanding the influence of these factors in a contextualized fashion is necessary to develop effective interventions that improve hypertension control and, in turn, reduce the incidence of comorbidities associated with it. The "*Implementation, Integration, and Institutionalization of a Community-Based Care Program to Reduce Cardiovascular Risk in Santander*" (RE-HOPE) study aimed to implement, integrate, and institutionalize a community-based care strategy to reduce cardiovascular disease risk among people with hypertension in

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Santander, Colombia. This study aimed to describe rates of hypertension control - with the goal of contributing to the local monitoring of the WHO target - and to answer: what are the individual-level factors associated with uncontrolled hypertension in Santander, Colombia?

MATERIALS AND METHODS

Study Design, setting, and participants

The present is a cross-sectional analysis of baseline data from the RE-HOPE study. The RE-HOPE is a sequential mixed-methods design study developed in primary healthcare centers in Santander, Colombia. RE-HOPE is composed of two phases: an initial qualitative phase to identify barriers and facilitators to hypertension control, informing the design of the intervention, followed by an intervention phase aimed at assessing the effectiveness of the implemented strategy. Adults aged 18 years and older with hypertension, selected by convenience sampling, were included. Hypertension was defined as: a) newly diagnosed hypertension (blood pressure average after three measurements ≥140/90 mmHg); or b) uncontrolled hypertension (blood pressure ≥140/90 mmHg in participants with a previous diagnosis of hypertension or who are currently on antihypertensive treatment); or c) known hypertensive participants and/or taking medication but not regularly attending cardiovascular risk management programs at primary care centers. Exclusion criteria included history of cardiovascular disease (e.g., coronary artery disease, myocardial infarction, heart failure, arrhythmias, cardiomyopathies, peripheral arterial disease, stroke or transient ischemic attack, diabetes mellitus with macrovascular complications, or history of invasive cardiovascular procedures), chronic kidney disease requiring dialysis, intolerance to antihypertensive medications, and resistant hypertension (defined as uncontrolled hypertension despite concurrent use of three antihypertensive medications of different classes at optimal doses) [17] or self-reported secondary hypertension. For the intervention phase, a total of 2 300 participants with uncontrolled hypertension were expected to be included. Enrollment took place from May to December 2023 across 12 municipalities, distributed across 6 provinces in the Santander department, Colombia. This report follows the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines [18].

Recruitment, variables, and data collection

Trained nursing assistants, supported by community leaders, conducted an active search in the community using a standardized screening protocol. During the screening phase, informed consent was obtained, eligibility criteria were verified, and blood pressure was measured. Participants with systolic blood pressure (SBP) ≥ 140 mmHg and/or diastolic blood pressure (DBP) ≥ 90 mmHg were scheduled for a verification visit within the next 7 days and were referred to their assigned primary care provider, using a standardized referral form. During the verification visit, blood pressure was measured again. Each measurement consisted of three repeated readings at one- to two-minute intervals. Measurements were conducted with the participant seated, at rest for at least five minutes, with their back supported, feet flat on the floor, and the arm uncovered and supported at heart level, in accordance with the International Society of Hypertension (ISH) guidelines [17]. Additionally, sociodemographic characteristics and information about comorbidities, medication intake, hypertension awareness, and lifestyle were collected through a standardized structured questionnaire. Anthropometrics, including weight, height, waist circumference, and hip circumference, were measured according to international protocols. Body mass index (BMI) was calculated as weight in kilograms divided by the square of height in meters (kg/m^2). Obesity was defined as a BMI ≥ 30 kg/m^2. Abdominal obesity was defined following IDF criteria as ≥90 cm for men and ≥80 cm for women [19]. Participants without a history of hypertension and blood pressure levels <140/90 mmHg during the verification visit were excluded. All participants provided written informed consent prior to enrollment. The data collection protocol was approved by the IRB of the Universidad de Santander (Act No. 018, dated August 29, 2022). The study was funded by Colombia's General System of Royalties BPIN code: 2020000100447

Statistical analysis

Descriptive statistics were presented to summarize baseline characteristics. For numerical variables with a normal distribution, the mean and standard deviation (SD) were estimated, while for non-normally distributed variables, the median and interquartile range (IQR) were calculated. Categorical variables were presented as counts and proportions. Mixed-effects logistic regression models were developed to evaluate the factors associated with uncontrolled hypertension. Models were stratified by sex and adjusted by potential confounders, such as age, diabetes, obesity, medication intake, physical activity, and smoking. Missing values were imputed using predictive mean matching via the mice R package (version 3.16.0). All statistical analyses were performed at a statistical significance level (α) of 0.05. The analysis was performed using the R software (free version 4.3.3).

RESULTS

Through active door-to-door screening and health screening events, 10 033 participants were invited to participate in the study. Of those, 9 806 participants accepted to participate in the screening and signed informed consent forms. Of those, 6 183 participants were excluded because they didn't meet the inclusion criteria (n = 5 951) or refused home visits (n = 232). Following the verification visit, 142 participants without a history of hypertension were excluded because their blood pressure values fell within the normal range. This led to a final study baseline sample of 3 481 participants (Fig. 1). The baseline characteristics, stratified by participants enrolled in the study and those not included, are presented in Table 1. The included participants had a lower proportion of women (62 vs. 66%), were older (60.2 vs. 52.7 years), and had higher SBP (145.7 vs. 117.9 mmHg) and DBP (86.3 vs. 74.8 mmHg) levels compared to participants excluded.

Table 2 shows the baseline characteristics of study participants. Among participants with hypertension, 40% were from the Metropolitan Area of Bucaramanga, followed by 23% from Yariguies Province. The mean SBP was 144 ± 16 mmHg, and the mean DBP was 86 ± 10 mmHg. The stratified analyses by age groups show that participants aged <60 years had lower SBP (141 vs. 147 mmHg; $p < 0.001$) but higher DBP (87 vs. 84 mmHg; $p < 0.001$). Additionally, low physical activity was higher among participants <60 years (93.5 vs. 84.0%) compared to those aged 60 years or older. Anthropometric measures, comorbidities, and lifestyle characteristics not described above are presented in Table 2.

Despite high rates of awareness (81%) and medication use (76%), only 26% of participants had controlled blood pressure (< 140/90 mmHg), and only 9.6% met the more stringent target of <130/80 mmHg (Table 3). However, within participants who were taking antihypertensive medication, hypertension control, defined as blood pressure values of <140/90 mmHg, was achieved by only 26%, with the highest rate in Garcia Rovira (52%) and the lowest in Yariguies (5.2%). Additionally, when the hypertension control was defined as blood pressure values < 130/80 mmHg, the prevalence of control was 9.6%, with higher rates in Garcia Rovira (29%) and lower rates in Vélez (0%).

Multivariable models showed that being a woman (OR = 0.62; 95%CI = 0.52-0.75) and having a history of diabetes (OR = 0.72; 95%CI = 0.58-0.88) were associated with lower likelihood of uncontrolled hypertension; while age ≥60 years (OR = 2.19; 95% CI = 1.84-2.61), being physically active (OR = 1.50; 95%CI = 1.09-2.06) and smoking (OR = 2.48; 95%CI = 1.54-4.00) increased the likelihood of uncontrolled hypertension. When stratifying by sex, both women and men aged ≥60 years (OR = 2.29; 95%CI = 1.85-2.83 and OR = 1.99; 95%CI = 1.47-2.70, respectively) and active smoking (OR = 3.14; 95%CI = 1.28-7.70 and OR = 2.14; 95%CI = 1.22-3.76, respectively) increased the likelihood of uncontrolled hypertension; conversely, diabetes status was associated with a better hypertension control incidence in both men and women (OR = 0.74; 95%CI = 0.58-0.96 and OR = 0.68; 95%CI = 0.47-0.98, respectively). Other associations are shown in Table 4.

DISCUSSION

This study reveals a gap between treatment and control, with only one-quarter of patients achieving BP targets. Despite a high

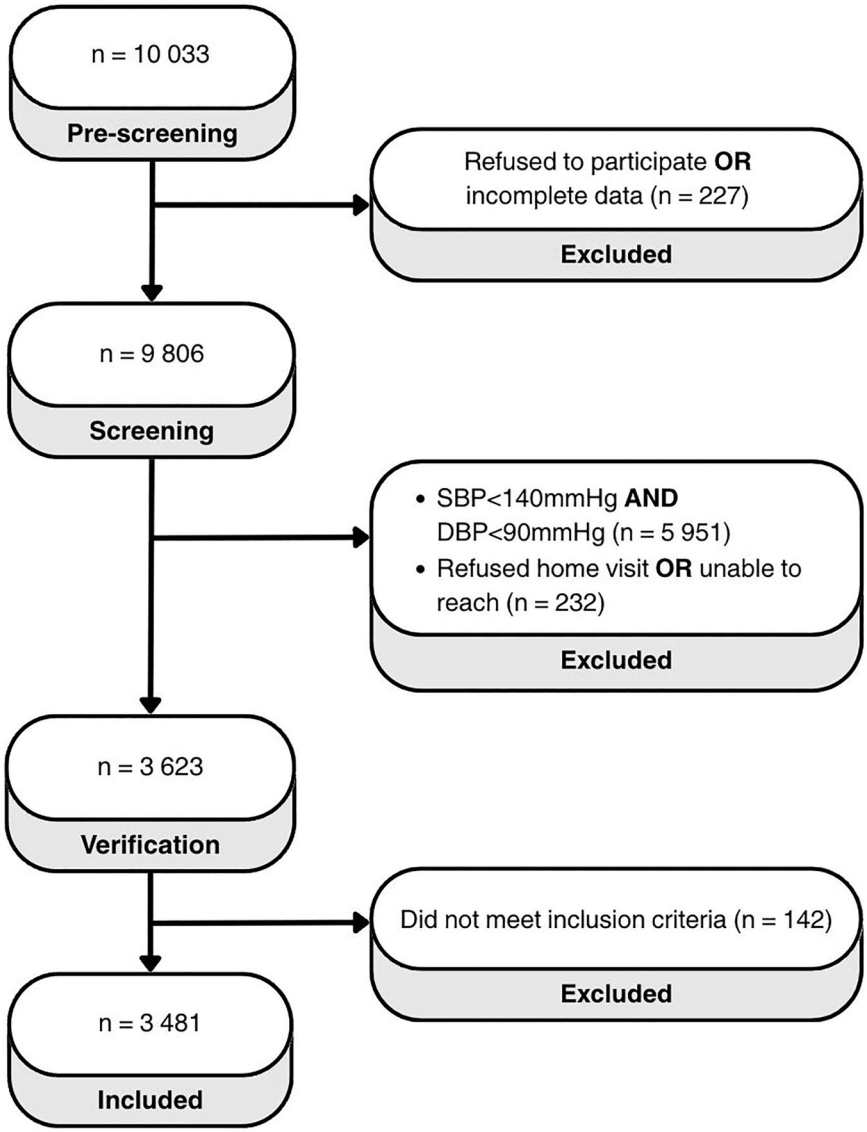

Fig. 1 Flowchart of the sample selection process. The RE-HOPE study.

frequency of awareness (81%) and medication intake (76%), only 26% of participants had controlled hypertension (< 140/90 mmHg). The results are worrisome, considering that, according to the 2024 European Society of Cardiology hypertension guidelines, blood pressure values should be targeted below 130/80 mmHg or "as low as reasonably achievable" [15]. We found that only 9.6% of the participants on medication met this control target. It should be noted that this more stringent target may not be applicable to all participants, particularly those aged 65 years and older. Our results showed that the proportion of participants with controlled blood pressure (< 140/90 mmHg) was higher compared to the PURE-Colombia cohort study, which demonstrated that among participants with hypertension receiving medication, only 17.3% were controlled [7]. However, RE-HOPE is not a population study, and it used convenience sampling, which limits comparability across studies. Furthermore, the PURE report was based on baseline data collected between 2005 and 2009, and several efforts and initiatives have been undertaken in recent decades to improve hypertension control [20]. Conversely, the rates of hypertension control reported in the present study are lower than those reported in other observational studies from Latin American countries; for instance, lower than those observed in Chile (44.3%), Uruguay (44.6%), and Perú (62.1%) [3, 21].

Notably, a higher percentage of control was found in the Metropolitan Area of Bucaramanga (the capital city) compared to peripheral and rural provinces. In Colombia, access to both education and income is generally lower in rural areas, and a direct association between education attainment and hypertension control has been demonstrated [7]. Our findings may reflect access barriers related to sociodemographic factors (such as low educational attainment and income), limited social support networks, difficulties in medication distribution, and logistical challenges—including commuting and transportation to healthcare facilities—which are particularly described in rural areas.

We found a substantial proportion of participants (22%) were undetected by the healthcare system, highlighting gaps in screening and diagnosis. This finding aligns with global data suggesting that a significant number of participants with hypertension remain undiagnosed and untreated [22]. Effective management and prevention of complications depend on timely diagnosis and intervention. Screening programs can significantly

Table 1. Blood pressure-related characteristics at the screening stage.

	Met selection criteria			
Characteristic	**Yes $n = 3\ 623$**	**No $n = 6\ 183$**	**Overall $n = 9\ 806$**	***p*-value**
Female – no. (%)	2 248 (62)	3 136 (66)	5 384 (64)	**<0.001**
Age (mean; SD)	60.2 ± 13.7	52.7 ± 18.7	56.0 ± 17.1	**<0.001**
SBP>140 mmHg – no. (%)	2 572 (71)	160 (4.1)	2 732 (36)	**<0.001**
DBP > 90 mmHg – no. (%)	1 353 (37)	245 (6.2)	1 598 (21)	**<0.001**
Self-reported hypertension – no. (%)	2 740 (76)	725 (15)	3 465 (42)	**<0.001**
Antihypertensive medication – no. (%)	2 569 (71)	694 (15)	3 263 (39)	**<0.001**
SBP (mean; SD)	145.7 ± 15.6	117.9 ± 13.9	131.2 ± 20.2	**<0.001**
DBP (mean; SD)	86.3 ± 10.3	74.8 ± 9.2	80.3 ± 11.3	**<0.001**

SD standard deviation, *SBP* systolic blood pressure, *DBP* diastolic blood pressure.

Table 2. Participants baseline characteristics.

Characteristic	**Age < 60 $n = 1\ 730$**	**Age ≥ 60 $n = 1\ 751$**	**Overall $n = 3\ 481$**	***p*-value**
Sociodemographic characteristics				
Sex female – no. (%)	1 102 (64)	1 060 (61)	2 162 (62)	0.067
Age (mean; SD)	50 ± 8	71 ± 7	61 ± 13	**<0.001**
Province – no. (%)				
Metropolitan Area of Bucaramanga	788 (46)	612 (35)	1 400 (40)	**<0.001**
Comunera	174 (10)	237 (14)	411 (12)	
Garcia Rovira	185 (11)	244 (14)	429 (12)	
Guanentá	158 (9.1)	206 (12)	364 (10)	
Vélez	35 (2.0)	41 (2.3)	76 (2.2)	
Yariguies	390 (23)	411 (23)	801 (23)	
Blood pressure levels				
SBP (mean; SD)	141 ± 16	147 ± 16	144 ± 16	**<0.001**
DBP (mean; SD)	87 ± 10	84 ± 10	86 ± 10	**<0.001**
Anthropometric measurements				
Waist circumference (mean; SD)	99 ± 13	96 ± 11	97 ± 12	**<0.001**
Hip circumference (mean; SD)	107 ± 12	102 ± 11	104 ± 11	**<0.001**
Weight (mean; SD)	76 ± 15	67 ± 12	72 ± 15	**<0.001**
Height (mean; SD)	1.62 ± 0.09	1.59 ± 0.09	1.60 ± 0.09	**<0.001**
BMI (mean; SD)	29.0 ± 5.0	26.5 ± 4.3	27.8 ± 4.8	**<0.001**
WHR (mean; SD)	0.93 ± 0.08	0.94 ± 0.08	0.93 ± 0.08	**<0.001**
Comorbidities				
Diabetes – no. (%)	303 (18)	385 (22)	688 (20)	**0.002**
Obesity by BMI – no. (%)	639 (37)	319 (18)	958 (28)	**<0.001**
Abdominal obesity – no. (%)	1 459 (84)	1 382 (79)	2 841 (82)	**<0.001**
Lifestyle				
Moderate-to-vigorous – no. (%)	112 (6.5)	277 (16)	389 (11)	**<0.001**
Smoking – no. (%)	90 (5.2)	84 (4.8)	174 (5)	0.6

SD Standard deviation, *SBP* Systolic blood pressure, *DBP* diastolic blood pressure, *BMI* body mass index, *WHR* waist to hip ratio.

reduce the incidence of cardiovascular events by identifying and managing hypertension early. However, barriers such as lack of awareness, insufficient healthcare infrastructure, and socioeconomic factors often hinder effective screening, particularly in LMICs [23]. Addressing these barriers through targeted public health initiatives and policy changes is crucial for improving hypertension control and reducing the global burden of CVD. This is particularly relevant in Latin America, where limited population-based strategies are available for addressing hypertension.

The most commonly adopted strategy is the use of standardized treatment guidelines [4]; however, the adherence of health team to those guidelines is far to be optimal, demonstrating the implementation gaps existing between hypertension identification, and achieving blood pressure control and underscores the importance of comprehensive management strategies that include patient education, lifestyle modifications, considerations of social determinants, and appropriate medication [24]. Latin American countries have implemented the HEARTS program, a

Table 3. Awareness and hypertension control by province.

Characteristic	Metropolitan Area of Bucaramanga $n = 1\,400$	Comunera $n = 411$	Garcia Rovira $n = 429$	Guanentá $n = 364$	Vélez $n = 76$	Yariguíes $n = 801$	Overall $n = 3\,481$	p-value
Awareness								
Self-reported hypertension – no. (%)	1 115 (80)	347 (84)	378 (88)	325 (89)	57 (75)	589 (74)	2 811 (81)	**<0.001**
Antihypertensive medication – no. (%)	1 008 (72)	352 (86)	379 (88)	324 (89)	46 (61)	526 (66)	2 635 (76)	**<0.001**
Hypertension control within participants under medication								
BP < 130/80 mmHg – no. (%)	140 (10)	30 (7.3)	125 (29)	26 (7.1)	0 (0)	14 (1.7)	335 (9.6)	**<0.001**
BP < 140/90 mmHg – no. (%)	498 (36)	77 (19)	221 (52)	54 (15)	26 (34)	42 (5.2)	918 (26)	**<0.001**

BP blood pressure.

community-based initiative designed to enhance hypertension control and its associated risk factors [25]. This strategy involves collaborative work to modify health services at the primary care level, thereby increasing hypertension control and cardiovascular risk [26]. However, several factors influence hypertension control, including patients' adherence to healthy lifestyles and medication, which is a major challenge. Factors such as belief, difficulties in accessing medication, unawareness, and lack of knowledge are barriers to adherence [23]. Interestingly, we found that people with hypertension under 60 years had worse anthropometric characteristics and lower levels of physical activity. The Latin American Society of Hypertension has identified obesity, diabetes, sedentarism, and alcohol intake as major risk factors for uncontrolled hypertension [3]. This highlights the importance of implementing screening and detection strategies not only for hypertension, but also for risk factors in earlier life stages.

Among participants in the RE-HOPE study, being a man was associated with uncontrolled hypertension. According to our results, a cross-sectional study involving 66 567 people with hypertension in Thailand found that men had a higher probability of uncontrolled hypertension compared with women (OR 1.16; 95% CI = 1.10 - 1.20) [27]. Likewise, an Iranian cross-sectional study in adults found that men had higher chances of uncontrolled hypertension (OR 2.12; 95%CI = 1.54 - 2.94) [28]. The multinational PURE cohort study demonstrated that in primary prevention, women had better lifestyles and hypertension control, contributing to a lower cardiovascular risk [29]. The above is likely related to differences in sociocultural behaviors that may delay access to health services, prevention, diagnosis, and early treatment, as well as affect adherence to lifestyle modifications and medication, rather than solely biological factors [30].

In addition, participants over 60 years and smokers had a higher likelihood of presenting uncontrolled hypertension. Hypertension control becomes increasingly challenging with age due to the progressive vascular stiffness associated with the aging process, which favors an increase in peripheral vascular resistance, leading to higher blood pressure levels. Furthermore, tobacco use can reduce the effect of antihypertensive medication through sympathomimetic effects. In this way, the World Heart Federation's tobacco control campaigns through multi-pronged initiatives, including public awareness programs, policy advocacy, and healthcare professional education, have contributed to decreased tobacco consumption in several target regions, particularly effective in LMICs, helping to shape stronger tobacco control policies and increased public understanding of smoking's cardiovascular risks [31].

We found that participants with diabetes had a lower likelihood of uncontrolled hypertension. While this finding may seem counterintuitive given the added cardiovascular risk, it is consistent with some prior studies [32]. Patients with diabetes are typically engaged in more frequent clinical monitoring and follow-up, creating more opportunities for hypertension management and medication titration [33]. Moreover, in the Colombian context, programs like the *Ruta Integral de Atención en Salud (RIAS)* for chronic diseases may create a more structured care pathway for patients with diabetes, inadvertently benefiting hypertension control through integrated management. However, the possibility of information bias cannot be ruled out. Our findings reveal a concerningly high prevalence of physical inactivity, with 89% of participants reporting no moderate-to-vigorous physical activity. The proportion of inactivity observed in this study far exceeds global age-standardized prevalence estimates and highlights the urgent need for lifestyle interventions to promote physical activity and support progress toward the 2025 global target of a 10% relative reduction in insufficient physical activity [20, 34]. Addressing these challenges requires a multifaceted approach, including patient education, enhanced social support, and accessible resources. In our context, successful strategies have shown

Table 4. Associated factors with uncontrolled hypertension in participants of the RE-HOPE study during baseline.

Variable	Women OR (95%CI)	*p*-value	Men OR (95%CI)	*p*-value	Overall	*p*-value
Female	-	-	-	-	0.62 (0.52 - 0.75)	**<0.001**
Age>60 years	2.29 (1.85 - 2.83)	**<0.001**	1.99 (1.47 - 2.70)	**<0.001**	2.19 (1.84 - 2.61)	**<0.001**
Diabetes	0.74 (0.58 - 0.96)	**0.021**	0.68 (0.47 - 0.98)	**0.037**	0.72 (0.58 - 0.88)	**0.002**
Obesity	0.93 (0.74 - 1.16)	0.503	1.12 (0.80 - 1.59)	0.507	0.99 (0.82 - 1.19)	0.909
Physical activity	1.43 (0.98 - 2.08)	0.064	1.72 (0.94 - 3.18)	0.081	1.50 (1.09 - 2.06)	**0.013**
Smoking	3.14 (1.28 - 7.70)	**0.013**	2.14 (1.22 - 3.76)	**0.008**	2.48 (1.54 - 4.00)	**<0.001**

Obesity: BMI ≥ 30 Kg/m^2; physical activity: at least 150 min per week of moderate-vigorous physical activity; smoking: current tobacco use.
BMI body mass index, *OR* odds ratio, 95%*CI* 95% confidence interval.

encouraging results. A randomized controlled multi-center trial in participants with high blood pressure (SBP between 130-140 mmHg) showed that three isometric training sessions per week (handgrip or wall squat) significantly decrease SBP (handgrip mean: 10.9 mmHg; $p = 0.003$ and wall squat mean: 14.7 mmHg; $p < 0.001$) compared with educational control [35].

This study has limitations. The cross-sectional design limits the ability to establish causality between identified factors and uncontrolled hypertension. Furthermore, the reliance on self-reported data for medication adherence and lifestyle habits may introduce reporting bias. The observed association between physical activity and uncontrolled hypertension appears counter-intuitive and is likely due to information bias or reverse causality. The study was conducted in the Andean region of Colombia, where Afro-descendant populations may be underrepresented, potentially limiting the generalizability of the findings to other regions or ethnic groups within the country. Notwithstanding, the study presents several notable strengths, particularly because it represents the baseline analysis of the RE-HOPE study, a longitudinal community-based implementation study. To our knowledge, this is one of the few studies in Colombia that characterizes participants with hypertension and examines factors affecting its control, thereby supporting the development of strategies that help reduce the disease burden. The value and potential impact of the research are underscored by its comprehensive data collection, large and diverse sample, identification of initial gaps, and its role in guiding the implementation of the RE-HOPE study strategies. These strengths ensure that the study provides valuable information to improve hypertension management and ultimately enhance health outcomes in the population.

CONCLUSION

At baseline, the RE-HOPE study found that among participants with hypertension, only 26% were well controlled. These findings underscore the urgent need to integrate early detection, lifestyle interventions, and tailored pharmacological strategies into primary care models across underserved regions, with a particular focus on adults under 60 years of age, given their higher levels of physical inactivity and poorer anthropometric indicators. These findings underscore the urgent need for tailored, place-based public health strategies that move beyond simple medication prescription to address the multifaceted barriers to control.

SUMMARY

What is known about topic

- In Latin America, reports of hypertension prevalence vary widely. Unlike other regions, its prevalence has increased in recent decades, while control rates remain low.
- Low control rates—particularly in low-income communities—have been consistently reported, despite the proven efficacy of both pharmacological and non-pharmacological interventions recommended by clinical guidelines.
- More evidence is needed to identify and describe factors associated with uncontrolled hypertension in order to guide effective interventions in low- and middle-income countries, such as Colombia.

What this study adds

- Despite a high frequency of awareness (81%) and medication intake (76%), only 26% of participants had controlled hypertension (< 140/90 mmHg) and 9.6% met the target below 130/80 mmHg.
- A substantial proportion of participants (22%) were undetected by the healthcare system, highlighting gaps in screening and diagnosis.
- Being male, over 60 years of age, and a current smoker are factors associated with uncontrolled hypertension and should be considered when identifying target groups for interventions aimed at improving hypertension control.

DATA AVAILABILITY

Individual-level data will not be shared because RE-HOPE is an ongoing implementation study. Requests for aggregated data will be considered on a case-by-case basis upon receipt of a reasonable request.

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AUTHOR CONTRIBUTIONS

JPL-L: Conceptualization, Methodology, Investigation, Supervision, Writing – Original Draft. YS-M: Conceptualization, Methodology, Writing – Original Draft. YG-C: Investigation, Methodology, Supervision, Writing – Review & Editing. DM-B: Investigation, Methodology, Formal analysis, Writing – Review & Editing. AC: Investigation, Methodology, Project administration, Writing – Review & Editing. CG: Investigation, Methodology, Writing – Review & Editing. ML-C: Investigation, Methodology, Writing – Review & Editing. JO: Conceptualization, Methodology, Investigation, Funding acquisition, Writing – Review & Editing. PL-J: Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing – review and editing

ETHICAL APPROVAL

The data collection protocol was approved by the IRB of the Universidad de Santander (Act No. 018, dated August 29, 2022). All methods were performed in accordance with relevant guidelines and regulations.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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